

MAHATMA GANDHI MEMORIAL COLLEGE

UDUPI – 576102



2025-2026

PROJECT SYNOPSIS

OF

“WORK ORDER MANAGEMENT SYSTEM”

To be carried out

At

**MANIPAL PAYMENT AND IDENTITY SOLUTIONS
LIMITED**

Submitted by

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1. Problem Statement

In many manufacturing and production environments, work orders are often managed manually through spreadsheets, paper documents, or unstructured communication between departments. This makes it difficult to track orders efficiently and coordinate activities between purchasing, planning, and production teams.

The current system lacks proper integration between different stages of the production process such as purchase order creation, production planning, and production reporting. As a result, organizations face problems such as delays in production, poor tracking of orders, lack of real-time updates, and difficulty in generating accurate reports.

Manual management of work orders also increases the chances of data entry errors and miscommunication between departments. Without a centralized system, it becomes challenging for management to monitor the progress of work orders and evaluate production performance.

Therefore, there is a need for a Work Order Management System that can streamline the entire workflow from purchase order creation to production reporting, improving efficiency and transparency within the organization.

2. Objectives

The main objectives of the Work Order Management System are:

- To manage and track work orders efficiently within the organization.
- To automate the purchase order and production planning process.
- To monitor production activities and maintain accurate records.
- To generate production reports for analysis and decision-making.
- To improve coordination between purchasing, planning, and production departments.

3. Scope

The system will provide a centralized platform for managing work orders throughout the production process. It will include modules for purchase order management, production planning, production tracking, and report generation.

The system will help managers and production teams monitor work order progress, manage resources efficiently, and improve overall productivity. It will also help reduce manual errors and enhance communication between departments.

4. Process Description

The Work Order Management System will consist of multiple modules that manage different stages of the production workflow.

1. Purchase Order

This module allows users to create and manage purchase orders for raw materials or components required for production. It records supplier details, order quantities, and delivery schedules.

2. Planning

The planning module is responsible for scheduling production activities based on the available resources and work orders. It ensures that materials, manpower, and machines are allocated efficiently.

3. Production

This module records the actual production process. It tracks work order execution, production quantities, and progress updates during manufacturing.

4. Production Report

The production report module generates reports that summarize production performance. It provides details such as completed work orders, production output, and pending tasks. These reports help management evaluate operational efficiency.

5. Resources and Limitations

Resources

Hardware

- Computer or Laptop
- Internet Connection
- Server for database storage

Software

- Programming Language: JavaScript (Node.js for backend, React.js for frontend)
- Database: MongoDB
- Development Tools: Visual Studio Code
- Web Technologies: HTML, CSS, Bootstrap, JavaScript, React.js

Data

- Work order details
- Purchase order data
- Production records

Limitations

- Limited time for full system development during the internship.
- Time constraints during the internship period.
- System scalability may be limited depending on infrastructure.

6. Conclusion

The Work Order Management System helps streamline the workflow from purchase order creation to production reporting. By automating the work order process, the system reduces manual effort and improves accuracy in production management. The proposed system enhances coordination between departments and provides better visibility in the production process. Managers can track work order progress and generate reports to make informed decisions. Future enhancements may include integration with inventory systems, real-time production monitoring, and advanced analytics for production optimization.

7. References

- [1] I. Sommerville, *Software Engineering*, 10th ed., Pearson Education, 2016.
- [2] R. Pressman, *Software Engineering: A Practitioner's Approach*, 8th ed., McGraw-Hill, 2015.
- [3] A. Dennis, B. Wixom, and R. Roth, *Systems Analysis and Design*, 6th ed., Wiley, 2012.